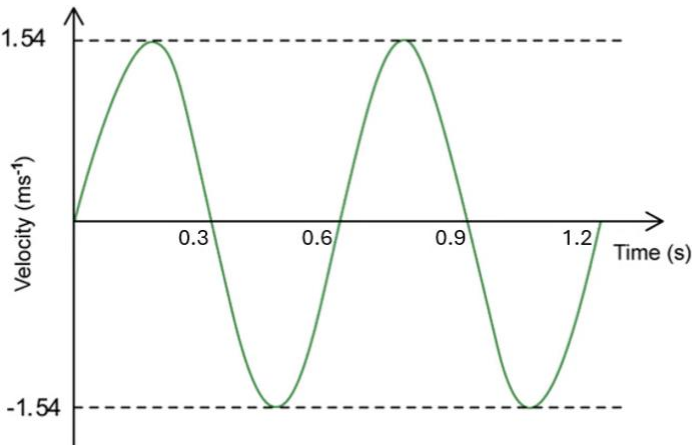


Q2	
(a)(i)	The mass passes through the equilibrium position P twice in a cycle, meaning that there are 100 complete cycles per minute. period, $T = 1/f = 60/100 = 0.600 \text{ s}$
(a)(ii)	At equilibrium position $x = 0$, the kinetic energy of the mass E_k is at its maximum. $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.600} = 10.47 \text{ rad s}^{-1}$ $E_k = \frac{1}{2} m \omega^2 x_o^2 = \frac{1}{2} (0.42) (10.47)^2 x_o^2$ $\text{Amplitude } x_o = \sqrt{\frac{2(0.500)}{0.42(10.47)^2}} = 0.147 \text{ m} = 14.7 \text{ cm} = 15 \text{ cm (2 s.f.)}$
(a)(iii)	$E_k = \frac{1}{2} m v^2 \Rightarrow v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2(0.500)}{0.42}} = 1.54 \text{ m s}^{-1}$  • Sinusoidal waveform • Maximum velocity of 1.54 m s^{-1} and time period of 0.60 s <i>*No information is given whether mass starts oscillating from its equilibrium position or at its amplitude – sine or cosine graph</i>
(b)(i)	No change
(b)(ii)	Frequency multiplied by $\sqrt{2}$.

